

Paper

An Empirical Study of VAE Bottleneck Design Guided by Intrinsic Dimension Estimation

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Abstract: We empirically examine a design guideline that uses the intrinsic dimension (ID) of training data to set the bottleneck size of a variational autoencoder (VAE). We first benchmark multiple ID estimators (PCA-90%, TwoNN, MLE, ESS, DANCo) on synthetic and real datasets and select reliable candidates for image data—TwoNN and MLE. We then compare the estimated ID with the practically optimal latent dimensionality of VAEs, identified by two complementary indicators: the number of Active Units (AU) and input–reconstruction similarity measured by linear CKA. On MNIST (single-class and all-class), we observe $\hat{d}_z^* \approx \hat{d}_{\text{ID}}$ or $\hat{d}_z^* \lesssim \hat{d}_{\text{ID}}$. We are extending experiments to Fashion-MNIST and CelebA to assess generality. This paper reports current results and discusses the usefulness and limitations of the guideline as well as future directions.

Key Words: Intrinsic Dimension, TwoNN, MLE, VAE, Active Units, CKA, Rate–Distortion

1. Introduction

Deep generative models, particularly variational autoencoders (VAEs), extract essential features from high-dimensional data such as images, yielding compact latent representations that have proved effective for data generation, compression, and anomaly detection. A key open issue is architectural design, especially the size of the bottleneck layer, which is still often chosen via heuristics or costly sweeps.

Our premise is the manifold hypothesis: high-dimensional observations lie near a lower-dimensional manifold. If the intrinsic dimension (ID) of the data can be estimated, it may serve as a principled guideline (lower/upper bounds) for the VAE latent dimensionality [1].

This work empirically validates the approach of leveraging the estimated $\hat{d}_{\text{ID}}$ to guide the design of the VAE bottleneck. We first objectively evaluate multiple ID estimators and identify practical ones. We then compare $\hat{d}_{\text{ID}}$ with the practically optimal VAE latent size $\hat{d}_z^*$, which we identify from two indicators: representation similarity (CKA) and the number of Active Units (AU).

Our contributions are threefold: (1) a coherent evaluation pipeline that links ID estimation to VAE design; (2) a cross-dataset analysis that reveals quantitative relations between $\hat{d}_{\text{ID}}$ and $\hat{d}_z^*$ (e.g.,



$\hat{d}_z^* \lesssim \hat{d}_{\text{ID}}$) and their dependence on data complexity; and (3) a reduction of trial-and-error toward a more principled design process.

2. Related Work

This study resides at the intersection of *intrinsic dimension estimation*, *architecture design for generative models*, and *quantitative evaluation of deep representations*.

2.1 Manifold Hypothesis and Intrinsic Dimension Estimation

The manifold hypothesis underpins dimensionality reduction. ID estimators include the maximum-likelihood approach based on extreme-value theory of neighbor distances (MLE) [2] and TwoNN based on the ratio of first- and second-nearest-neighbor distances [3]. While efficient, they can be sensitive to noise, curvature, and sample size. Robustification has been pursued through methods such as DANCo, which couples angle and norm concentration [4], and ESS based on simplex skewness. We select estimators by practical criteria (accuracy, stability, computational cost).

2.2 Choosing the Latent Dimensionality in Generative Models

In VAEs [5], too-small latent spaces underfit, while excessively large ones may cause training instabilities or posterior collapse [6][7]. Prior work often relies on β -VAE regularization [8] or validation-based searches. Our approach is to narrow the search space for the latent dimensionality using the data’s estimated ID.

2.3 Quantifying Deep Representations

To quantify the *effective use* of latent dimensions and information preservation, we employ Active Units (AU) [9] and Centered Kernel Alignment (CKA) [10].

3. Proposed Framework

We aim to empirically validate the use of the data’s intrinsic dimension as a prior for setting the VAE bottleneck size. Figure 3.1 outlines a two-stage pipeline.

Stage 1: Selecting ID estimators. We assess multiple ID estimators on synthetic datasets with known ground-truth dimension to secure reliable downstream analysis. Based on accuracy, stability, and runtime, we select estimators for application to real image data.

Stage 2: Relating ID to the optimal latent size. Using the estimators selected in Stage 1, we estimate $\hat{d}_{\text{ID}}$ for MNIST (all classes and single classes). In parallel we train VAEs while sweeping d_z and identify $\hat{d}_z^*$ from AU and CKA. We then analyze the relation between $\hat{d}_{\text{ID}}$ and $\hat{d}_z^*$ across datasets.

3.1 Estimating the Intrinsic Dimension ($\hat{d}_{\text{ID}}$)

Given training data X , we apply Stage-1 winners to obtain $\hat{d}_{\text{ID}}$. Based on Section 4, we adopt TwoNN and MLE as the primary estimators for image data with nonlinear manifold structure. Discrepancies between the two are interpreted as a confidence interval for the ID.

3.2 Identifying the Optimal Latent Size ($\hat{d}_z^*$)

Because VAE performance is multi-faceted, we combine two indicators and identify the saturation/plateau region as $\hat{d}_z^*$.

Active Units (AU) For latent $z = (z_1, \dots, z_{d_z})$, define

$$\text{Var}_x [\mathbb{E}_{q_\phi(z_i|x)}[z_i]],$$

and count dimensions with variance above a threshold δ (conventionally $\delta \approx 10^{-2}$) as “active” [9]. We estimate AU post-training from batch statistics.

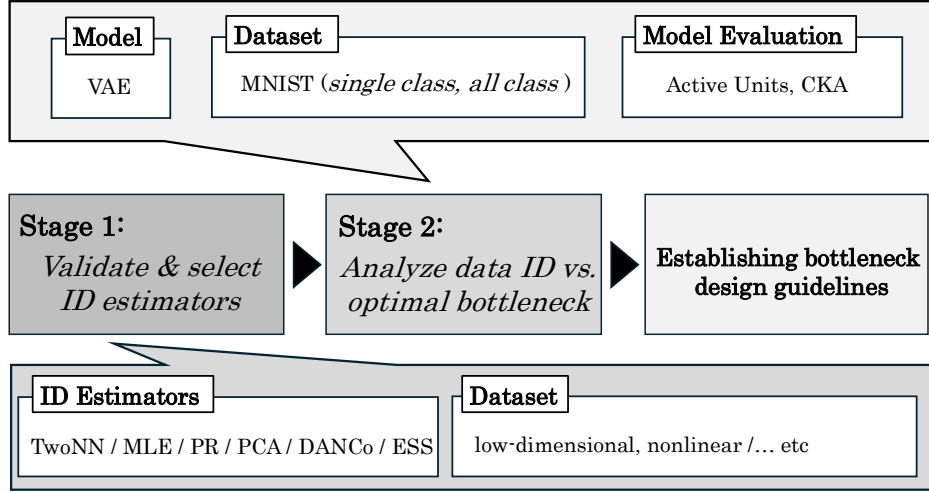


Fig. 3.1: Proposed pipeline: Stage 1 validates and selects ID estimators; Stage 2 relates the data ID ($\hat{d}_{\text{ID}}$) to the VAE’s optimal latent size ($\hat{d}_z^*$).

Centered Kernel Alignment (CKA) Linear CKA between two representations (matrices) is

$$\text{CKA}(K, L) = \frac{\|X^\top \hat{X}\|_F^2}{\|X^\top X\|_F \|\hat{X}^\top \hat{X}\|_F},$$

with $K=XX^\top$, $L=\hat{X}\hat{X}^\top$ [10]. We compute CKA between flattened, centered input X and reconstruction $\hat{X}$ and use the saturation point along d_z as a sufficiency marker for information preservation.

We take the overlap of AU- and CKA-derived ranges, accounting for variability, as the practical optimum $\hat{d}_z^*$.

3.3 Expected Relationship

Under the manifold hypothesis, d_{ID} is a lower bound for lossless representation. However, nonlinear decoding and KL regularization may promote sparse/compact use of capacity, suggesting $\hat{d}_z^*$ is comparable to or slightly smaller than $\hat{d}_{\text{ID}}$, i.e., $\hat{d}_z^* \lesssim \hat{d}_{\text{ID}}$. We test this across datasets.

4. Stage 1: Quantitative Evaluation of ID Estimators

4.1 Experimental Setup

We employ two groups of datasets: synthetic sets with known d_{true} for estimator evaluation (Stage 1), and image datasets for Stage 2. Synthetic sets include classic low-dimensional manifolds (Swiss roll, S-curve, two moons, concentric circles, Klein bottle), higher-dimensional families (unit spheres S^d , tori T^k , Stiefel $V_k(\mathbb{R}^n)$, Grassmann $G(k, n)$, orthogonal groups $\text{SO}(n)$, SPD matrices $\text{SPD}(n)$), linear subspaces embedded in higher-dimensional ambient spaces, and i.i.d. Gaussian baselines. We sample 20 random seeds, using 10,000 points if $d_{\text{true}} < 80$ and 20,000 points otherwise. Distances are Euclidean; to avoid zero distances for nearest-neighbor methods (TwoNN/MLE/ESS/DANCo), we add machine-epsilon jitter on exact duplicates.

We compare TwoNN, MLE, Participation Ratio (PR), PCA-90, ESS, and DANCo. Below we outline each method (omitted here for brevity; see full text).

Scoring. We normalize three criteria to $[0, 1]$: accuracy (S_{acc} via median absolute error), stability (S_{stab} via MAD), and runtime cost (S_{cost} via within-dataset linear normalization). The overall score is $S_{\text{mean}}=(S_{\text{acc}}+S_{\text{stab}}+S_{\text{cost}})/3$. For k -NN-based estimators we set $k=20$ (TwoNN uses $k=2$ by definition). Outliers/NaNs are handled robustly by medians and MAD.

Table I: Benchmark datasets grouped by type

Group	Dataset	Ambient D	True manifold dim.
G1: Low-d, nonlinear	KleinBottle_R4	4	2
	SwissRoll_R3	3	2
	SCurve_R3	3	2
	ConcentricCircles_R2	2	1
	TwoMoons_R2	2	1
G2: Isotropic / quasi-linear	GaussianRandom_D100	100	100
	Sphere_S10_R11	11	10
	Sphere_S50_R51	51	50
	Torus_T50_R100	100	50
	LinearSubspace_k50_D200	200	50
G3: Algebraic constraints	SO_6	36	$\frac{6 \cdot 5}{2} = 15$
	SO_10	100	$\frac{10 \cdot 9}{2} = 45$
	SO_15	225	$\frac{15 \cdot 14}{2} = 105$
	SO_20	400	$\frac{20 \cdot 19}{2} = 190$
	Stiefel_V3_R10	30	$10 \cdot 3 - \frac{3 \cdot 4}{2} = 24$
	Stiefel_V5_R20	100	$20 \cdot 5 - \frac{5 \cdot 6}{2} = 85$
	Stiefel_V8_R30	240	$30 \cdot 8 - \frac{8 \cdot 9}{2} = 204$
	Grassmann_G3_R12	78	$3(12 - 3) = 27$
	Grassmann_G5_R20	210	$5(20 - 5) = 75$
	Grassmann_G10_R30	465	$10(30 - 10) = 200$
	SPD_20	210	$\frac{20 \cdot 21}{2} = 210$

Table II: Estimation accuracy (S_{acc})

rank	estimator	mean_S_acc
1	TwoNN	0.769
2	MLE@20	0.732
3	ESS@20	0.697
4	PCA90	0.492
5	PR	0.471
6	DANCo@20	0.364

Table III: Stability (S_{stab})

rank	estimator	mean_S_stab
1	PCA90	1.000
2	DANCo@20	0.999
3	PR	0.999
4	ESS@20	0.998
5	MLE@20	0.998
6	TwoNN	0.991

Table IV: Computation cost (S_{cost})

rank	estimator	mean_S_cost
1	PR	0.995
2	PCA90	0.991
3	TwoNN	0.842
4	MLE@20	0.799
5	ESS@20	0.762
6	DANCo@20	0.000

Table V: Overall score (S_{mean})

rank	estimator	S_mean
1	TwoNN	0.867
2	MLE@20	0.843
3	PCA90	0.828
4	PR	0.821
5	ESS@20	0.819
6	DANCo@20	0.454

5. Stage 2: Relating the VAE Bottleneck to the Intrinsic Dimension

Guided by Stage 1, we estimate $\hat{d}_{ID}$ using TwoNN and MLE, then relate it to the optimal VAE latent size. We report results for MNIST (single-class and all-class). The Variational Autoencoder (VAE) used in this study consists of fully connected neural networks for both the encoder and decoder, each with two hidden layers. The input images are flattened into 784-dimensional vectors, and the numbers of units in the hidden layers are set to 512 and 256, respectively. The ReLU function is used as the activation function, except for the output layer, which employs the sigmoid function. For optimization during training, the Adam optimizer is used with a learning rate of 1e-3 and a batch size of 128.

5.1 Estimating the ID of MNIST

From MNIST (each single class and all ten classes), we subsample 80% and estimate the ID using TwoNN and MLE. Results are in Table VI.

Dataset	TwoNN estimate	MLE estimate	ID range
All classes	14.78 ± 0.06	14.97 ± 0.01	[14, 15]
Class 0	13.58 ± 0.21	12.49 ± 0.03	[12, 13]
Class 1	11.43 ± 0.13	9.60 ± 0.03	[9, 11]
Class 2	13.72 ± 0.13	13.67 ± 0.04	[13, 13]
Class 3	14.57 ± 0.13	14.62 ± 0.04	[14, 14]
Class 4	12.96 ± 0.09	13.11 ± 0.05	[12, 13]
Class 5	14.29 ± 0.15	14.17 ± 0.06	[14, 14]
Class 6	12.58 ± 0.16	12.16 ± 0.04	[12, 12]
Class 7	11.95 ± 0.12	11.83 ± 0.04	[11, 12]
Class 8	14.86 ± 0.18	15.20 ± 0.03	[14, 15]
Class 9	12.76 ± 0.14	12.43 ± 0.03	[12, 12]

Table VI: Estimated intrinsic dimensions ($\hat{d}_{ID}$) for MNIST (single classes and all classes)

5.2 Identifying the Optimal VAE Latent Size

We train VAEs while sweeping $d_z \in \{2, \dots, 40\}$ and evaluate AU and CKA (Appendix B). For all classes, AU gradually saturates and CKA plateaus; jointly these indicate $\hat{d}_z^* \in [8, 12]$. For Class 0 we obtain [5, 10].

6. Discussion

Table VII summarizes the relation between $\hat{d}_{ID}$ and $\hat{d}_z^*$. Across datasets, our results support the hypothesis that $\hat{d}_z^* \leq \hat{d}_{ID}$, and often find the stronger relation $\hat{d}_z^* \approx \hat{d}_{ID}$. This suggests that, when trained effectively, VAEs tend to use a number of latent degrees of freedom close to the data’s intrinsic degrees of freedom. This is consistent with a Rate–Distortion view [11], where the ELBO’s KL regularization trades off against reconstruction, nudging toward the minimal sufficient rate. This rate–distortion tradeoff is closely related to the relationship between the perceptual quality of generated images and their fidelity to the original data (the perception–distortion tradeoff) [citation]. The results of this study suggest that the VAE optimizes this tradeoff through a single objective function—the ELBO—based on the intrinsic dimensionality, which reflects a data-dependent property [12]. On the other hand, as shown in Table VI, a slight variation can be observed in the relationship between the optimal latent dimension and the intrinsic dimension across different dataset classes. In particular, for digits such as “1” (Class 1) and “8” (Class 8), the optimal latent dimension was found to be significantly smaller than the intrinsic dimension ($\hat{d}_z^* \ll \hat{d}_{ID}$). This may be because the digit “1” is primarily composed of straight lines and thus exhibits lower shape diversity, making it representable with fewer degrees of freedom. Similarly, although the digit “8” has a strong topological constraint due to its two closed loops, the variation in writing styles is relatively limited. These results suggest that the inherent geometric and topological characteristics of each class may influence the relationship between latent and intrinsic dimensions. Consequently, this finding implies that considering not only a single ID value estimated from the entire dataset but also the local structural complexity within the data could lead to a more precise bottleneck design.

Dataset	ID range ($\hat{d}_{\text{ID}}$)	VAE optimum ($\hat{d}_z^*$)	Relation
MNIST (all classes)	[14, 15]	[8, 12]	$\hat{d}_{\text{ID}} \gtrsim \hat{d}_z^*$
MNIST (Class 0)	[12, 13]	[5, 10]	$\hat{d}_{\text{ID}} \gtrsim \hat{d}_z^*$
MNIST (Class 1)	[9, 11]	[4, 6]	$\hat{d}_{\text{ID}} \gg \hat{d}_z^*$
MNIST (Class 2)	[13, 13]	[6, 12]	$\hat{d}_{\text{ID}} \gtrsim \hat{d}_z^*$
MNIST (Class 3)	[14, 14]	[6, 12]	$\hat{d}_{\text{ID}} \gtrsim \hat{d}_z^*$
MNIST (Class 4)	[12, 13]	[6, 13]	$\hat{d}_{\text{ID}} \approx \hat{d}_z^*$
MNIST (Class 5)	[14, 14]	[6, 11]	$\hat{d}_{\text{ID}} \gtrsim \hat{d}_z^*$
MNIST (Class 6)	[12, 12]	[6, 10]	$\hat{d}_{\text{ID}} \gtrsim \hat{d}_z^*$
MNIST (Class 7)	[11, 12]	[6, 11]	$\hat{d}_{\text{ID}} \approx \hat{d}_z^*$
MNIST (Class 8)	[14, 15]	[5, 10]	$\hat{d}_{\text{ID}} \gg \hat{d}_z^*$
MNIST (Class 9)	[12, 12]	[6, 10]	$\hat{d}_{\text{ID}} \gtrsim \hat{d}_z^*$

Table VII: Comparison between estimated ID and optimal latent size (integer-truncated $\hat{d}_{\text{ID}}$)

7. Limitations and Future Work

We have focused on MNIST; it remains to test datasets of higher complexity (Fashion-MNIST, CelebA). The network architecture was fixed in this study; isolating the effects of decoder expressivity is an important direction for future work. TwoNN/MLE are not universally robust under extreme curvature, multi-scale structure, or strong noise; quantifying how ID estimation error perturbs the $\hat{d}_{\text{ID}}-\hat{d}_z^*$ relation is future work. Finally, weak decoders may require $\hat{d}_z^* > \hat{d}_{\text{ID}}$. Systematically studying the relationship between decoder capacity, optimal latent size, and ID is an important direction for future work.

8. Conclusion

We evaluated the validity of using intrinsic dimension estimates to guide the VAE bottleneck size. After benchmarking ID estimators and selecting TwoNN/MLE for image data, we compared $\hat{d}_{\text{ID}}$ with $\hat{d}_z^*$ defined via AU and CKA. On MNIST we observed a strong positive correspondence, often $\hat{d}_z^* \approx \hat{d}_{\text{ID}}$, indicating that ID offers a powerful prior for narrowing the search range of VAE latent dimensionality and reducing design trial-and-error.

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Conflicts of interest

The authors declare no competing interests.

Author contribution

All the authors contributed equally to the present work.

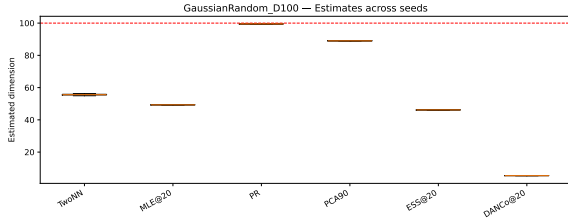
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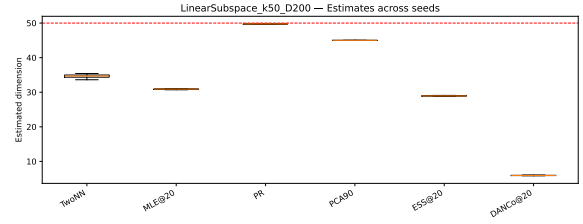
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Appendix

A. Appendix A: Stage 1 — Per-dataset estimation results (subset)

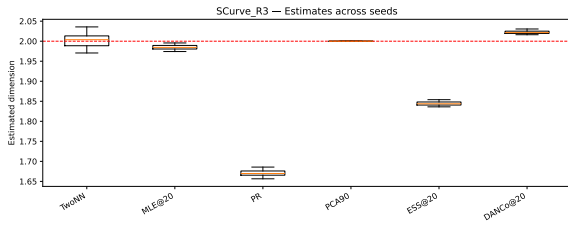


(a) GaussianRandom ($D=100$)

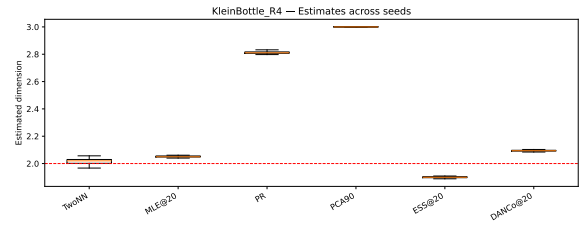


(b) LinearSubspace ($k=50, D=200$)

Fig. A-1: Distribution of estimated dimensions: baselines

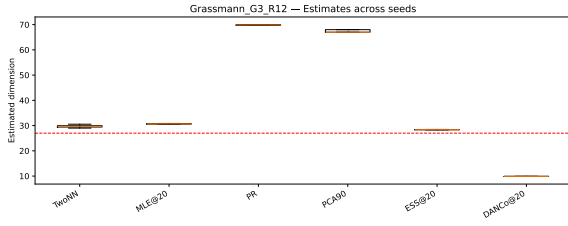


(a) S-curve ($\mathbb{R}^3$)

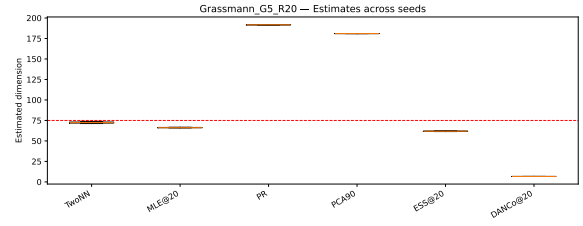


(b) Klein bottle ($\mathbb{R}^4$)

Fig. A-2: Distribution of estimated dimensions: nonlinear low-d manifolds

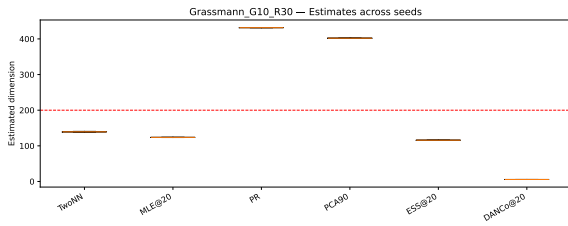


(a) $\text{Gr}(3, 12)$

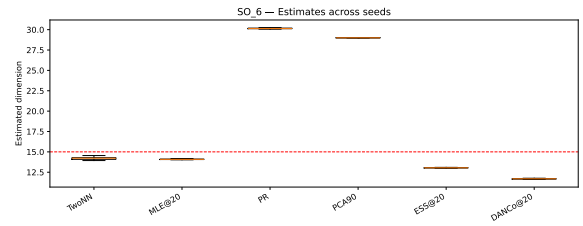


(b) $\text{Gr}(5, 20)$

Fig. A-3: Distribution of estimated dimensions: Grassmann (low-d settings)

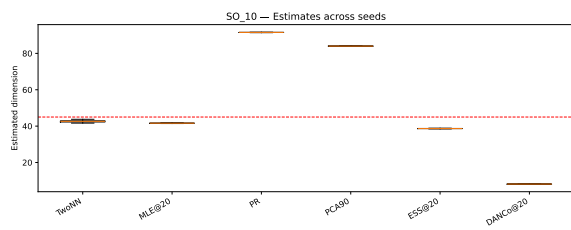


(a) $\text{Gr}(10, 30)$

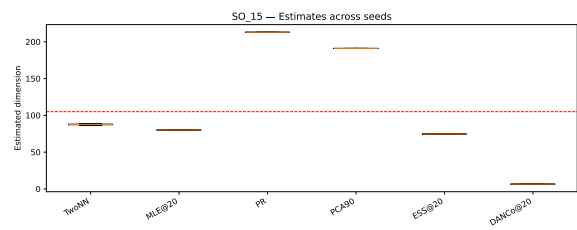


(b) $\text{SO}(6)$

Fig. A-4: Distribution of estimated dimensions: higher-d manifolds (I)



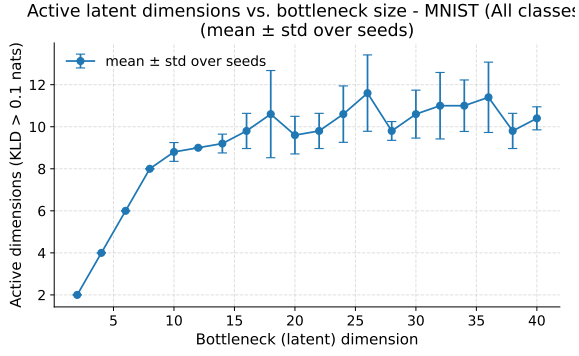
(a) SO(10)



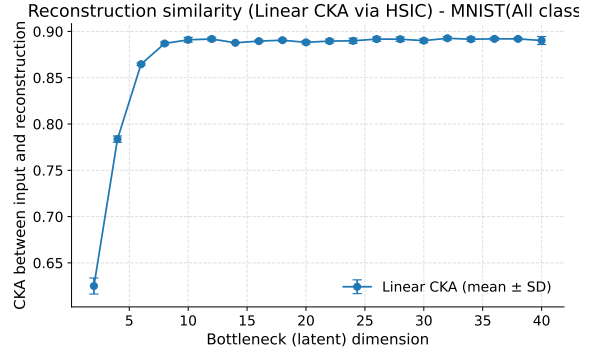
(b) SO(15)

Fig. A-5: Distribution of estimated dimensions: higher-d manifolds (II, SO family)

B. Appendix B: Stage 2 — Per-dataset VAE results

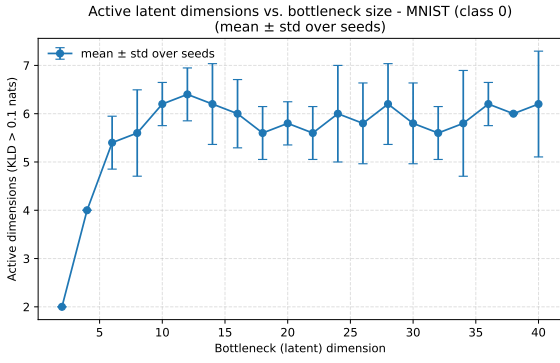


(a) Active latent dimensions

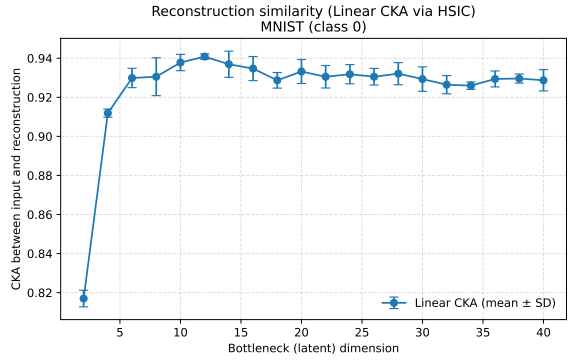


(b) Representation similarity (CKA)

Fig. B-1: MNIST (all classes): AU and CKA vs. d_z

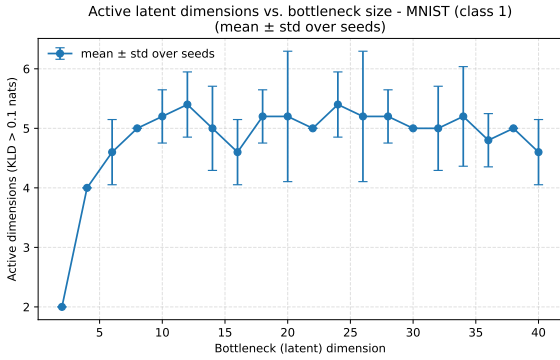


(a) Active latent dimensions

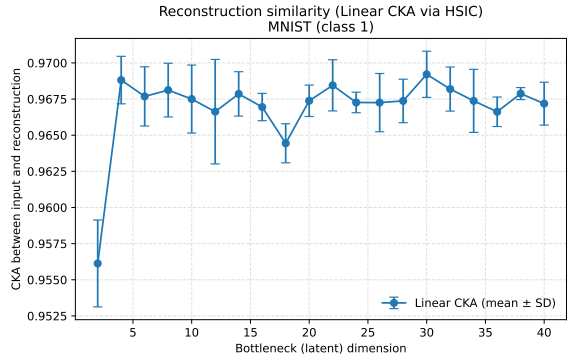


(b) Representation similarity (CKA)

Fig. B-2: MNIST (Class 0): AU and CKA vs. d_z

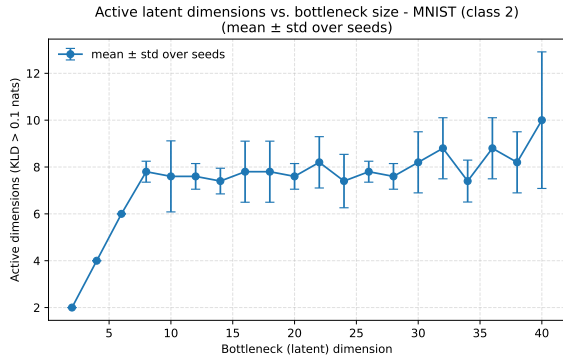


(a) Active latent dimensions

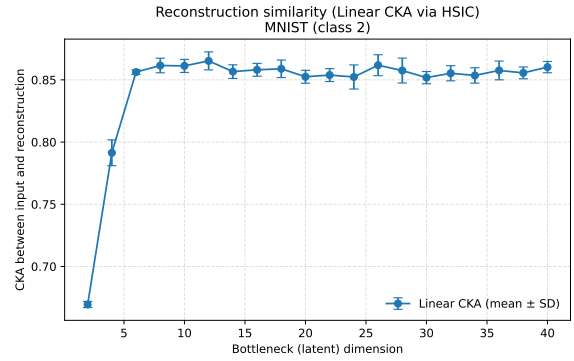


(b) Representation similarity (CKA)

Fig. B-3: MNIST (Class 1): AU and CKA vs. d_z

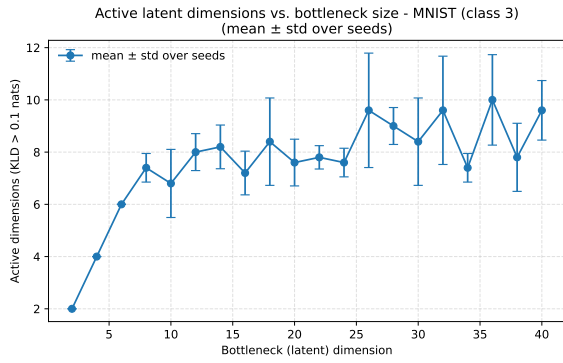


(a) Active latent dimensions

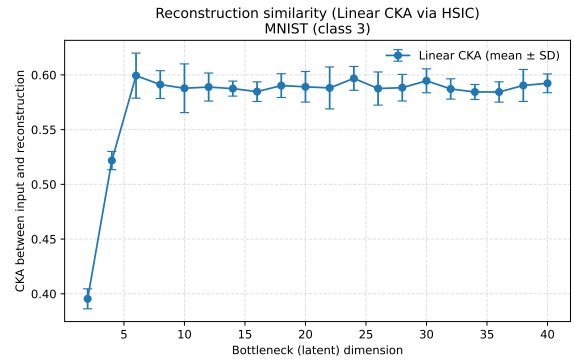


(b) Representation similarity (CKA)

Fig. B-4: MNIST (Class 2): AU and CKA vs. d_z

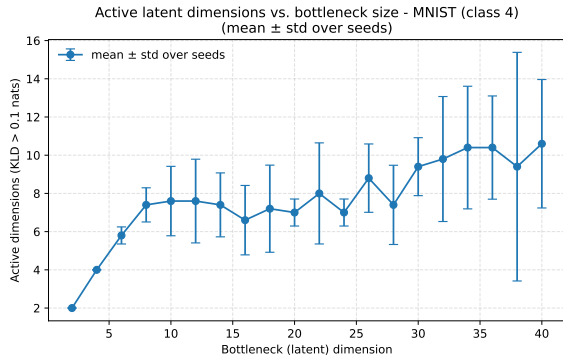


(a) Active latent dimensions

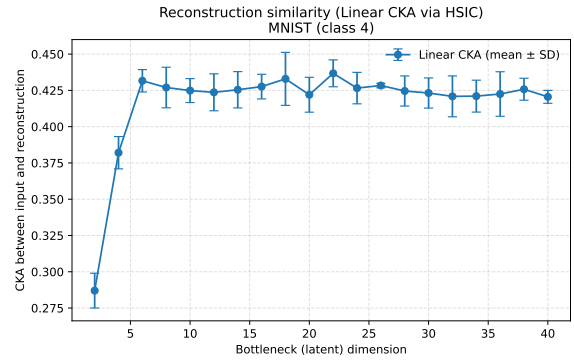


(b) Representation similarity (CKA)

Fig. B-5: MNIST (Class 3): AU and CKA vs. d_z

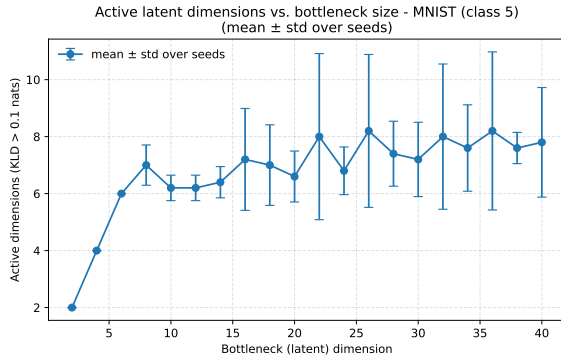


(a) Active latent dimensions

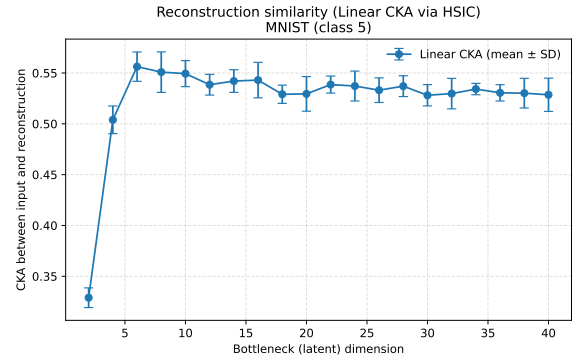


(b) Representation similarity (CKA)

Fig. B-6: MNIST (Class 4): AU and CKA vs. d_z

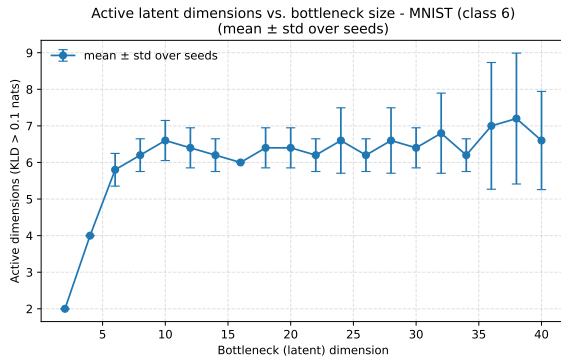


(a) Active latent dimensions

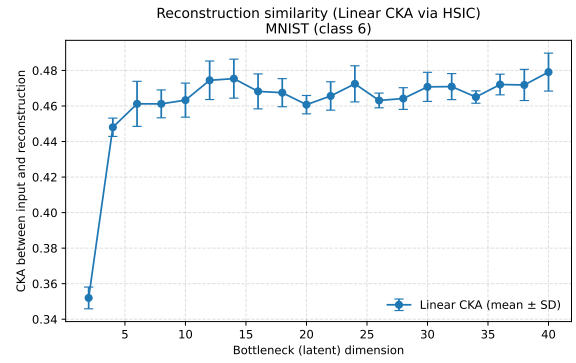


(b) Representation similarity (CKA)

Fig. B-7: MNIST (Class 5): AU and CKA vs. d_z

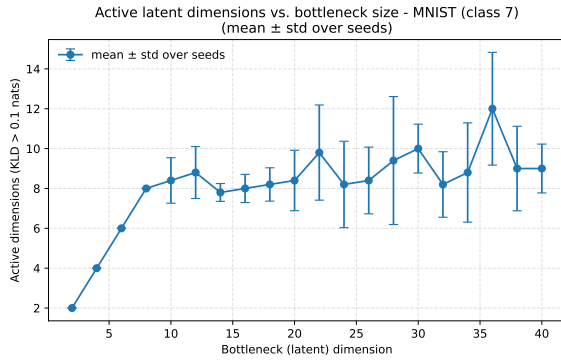


(a) Active latent dimensions

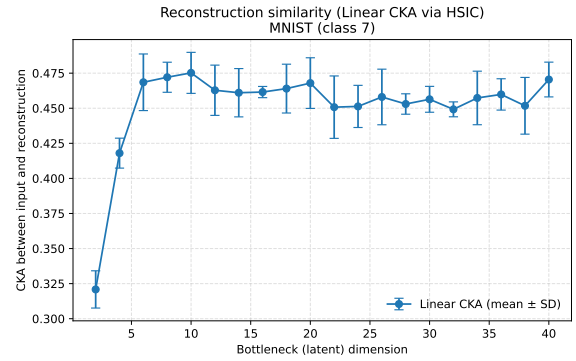


(b) Representation similarity (CKA)

Fig. B-8: MNIST (Class 6): AU and CKA vs. d_z

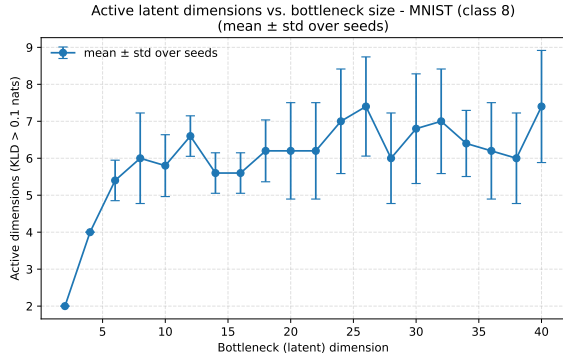


(a) Active latent dimensions

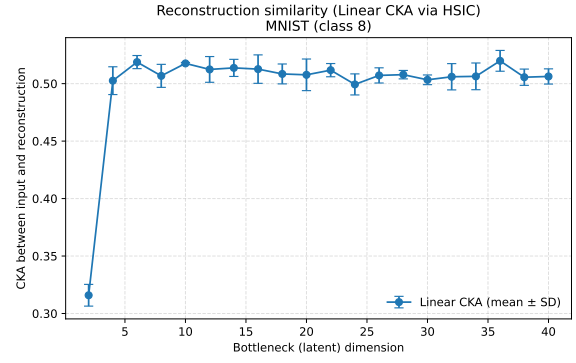


(b) Representation similarity (CKA)

Fig. B-9: MNIST (Class 7): AU and CKA vs. d_z

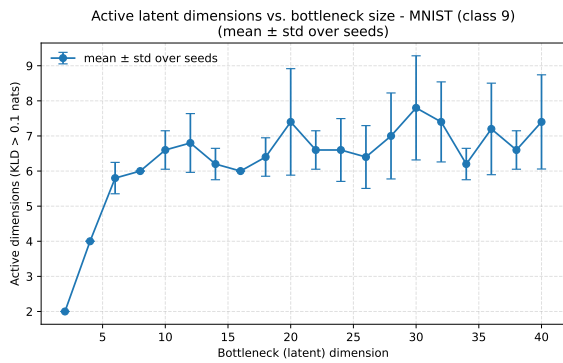


(a) Active latent dimensions

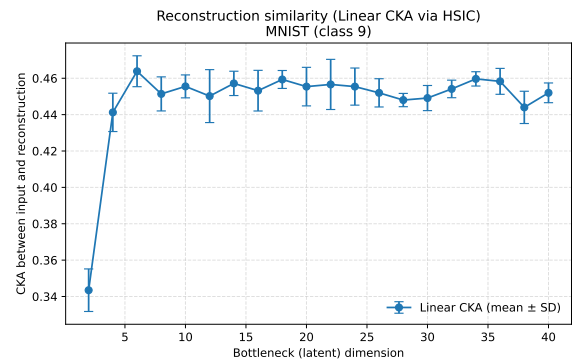


(b) Representation similarity (CKA)

Fig. B-10: MNIST (Class 8): AU and CKA vs. d_z



(a) Active latent dimensions



(b) Representation similarity (CKA)

Fig. B-11: MNIST (Class 9): AU and CKA vs. d_z